

AKADEMIKA

BCL-ASK

ASK MODULATION / DEMODULATION

KIT

EXPERIMENTAL MANUAL

.... A MESSAGE FROM



Today's system designers are faced with tomorrow's problems. DIGITAL COMMUNICATION is one of the important subject need to teach while learning electronics.

It is our vision to provide you with the product you need for training ensuring lasting reliability & quality.

OUR MOTTO;

- Light years ahead – refers to leadership.

As leaders in our industry in India, we are totally committed to servicing as the standard against which all are measured in the areas of

• Design • Quality • Value • Delivery • Support.

We are truly light years ahead of our competition in this area. That means that you our valued customers are guaranteed satisfaction.

As you will move through this manual you will quickly discover that we have complete, truly innovative & superior training products. We are so committed to quality that we back our products with a complete comprehensive warranty.

AKADEMIKA LAB SOLUTIONS

Sr. No. 15/8/1, Unit No.9, Kruti Industrial Estate, Opp.
Sangam Press, Kothrud, Pune, Maharashtra- 411038
INDIA

Contact No.:+91-9004904462

+91-7447438443

Email: lab@akademika.in

www.akademika.in

SAFETY RULES

Carefully follow the instructions contained in this handbook as they give important points on safety during the installation, use and maintenance. Keep this handbook always with you for any further help.

After the unpacking, set all accessories in order so that they are not lost and check the kit integrity. In particular, check that the kit is integral and shows no visible damage.

Before connecting the power supply to the kit, be sure that the jumpers and patch chords are connected correctly as per experiment.

In BCL-ASK kit, signals are to be monitored with an oscilloscope or with help of indicators as explained in the manual.

This kit must be employed only for the use for which it has been conceived, i.e. as educational equipment and must be used under the direct survey of expert personnel. Any other use is improper and so dangerous. The manufacturer cannot be considered responsible for eventual damages due to improper, wrong or unreasonable uses.

In case of any fault/bad operation, turn off the power supply and do not tamper the kit. In case servicing is required, contact the service center for technical assistance.

The kits are liable to malfunction/under-perform if they are not operated under standard environmental conditions of temperature and humidity.

WARRANTY

This kit is warranted against defects in workmanship and materials. Any failure due to defect in either workmanship or material should occur under normal use within a year from the original date of purchase, such failure will be corrected free of charge to the purchaser by repair or replacement of defective part or parts. When the failure is result of user's neglect, natural disaster or accident, we charge for repairs regardless of the warranty period. The warranty does not cover perishable items like connecting chords, Crystals, etc. and other imported items.

This warranty is subject to the following conditions and limitations. The warranty is void and inapplicable if the defective product is not brought or sent to our authorized service center or sales outlet within the warranty period. Defective product is, on Akademika Lab Solutions sole judgment. The defective product will be replaced with new one or repaired without charge or with charge.

In the warranty period if the service is needed, purchaser should get in touch with the service center or sales outlet. The purchaser should return the product to the service center or sales outlet at his or her sole expense. The loss and damage in transit will be outside the preview of this warranty. A returned product must be accompanied by a written description of the defects. Type and Model No. of kits has to be mentioned specifically. We return the product to the purchaser at our expense. In case that warranty does not cover the product on Akademika Lab Solutions judgment, we repair the product after obtaining prior permission from purchaser who receives an estimate statement about repairing charges. In such case, Akademika Lab Solutions bares the transporting expenses required to send back all the repaired products for the moment, and then repairs and transporting expenses will be charged against the purchaser by the statement of accounts.

When the authorized sales agents sell our products, they must notify the purchaser of the warranty contents, but have no rights to stretch the meaning of original warranty contents or offer additional warranty. Akademika Lab Solutions does not provide any other promise or suggestive warranty and hold no liability for the damage caused by negligence, abnormal use or natural disaster. Akademika Lab Solutions is not responsible for the damages even if it is notified of above dangers in advance as well.

For more special service or overall repairs, maintenance and up gradation of products, be sure to contact our service center or sales outlet.

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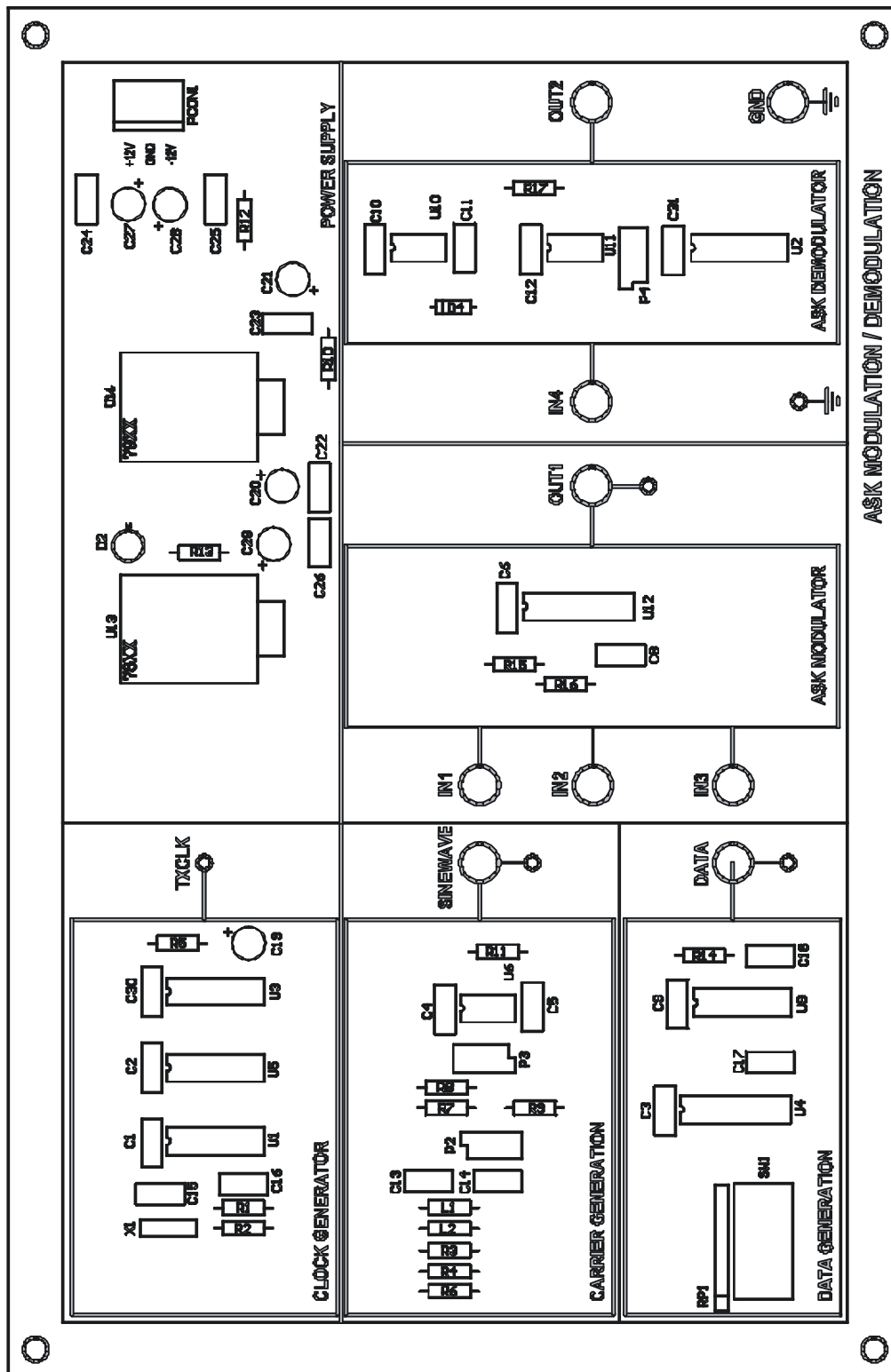
TECHNICAL SPECIFICATIONS

DATA SIMULATOR	:	Onboard 8-bit variable NRZ-L pattern.
CRYSTAL OSCILLATOR	:	2 MHz
DATA CLOCK	:	256 KHz.
ON BOARD CARRIER SINE WAVES	:	~ 1 MHz 4 Vp-p
CARRIER MODULATION	:	ASK.
CARRIER DEMODULATION	:	ASK
INTERMEDIATE SIGNAL	:	Provision for observing intermediate signals during demodulation.
INTER CONNECTION	:	2 mm banana socket.
POWER SUPPLY	:	+12V, -12V, GND.

Accessories:

BCL-ASK		Quantity
1	Short Link Patch Cords male to male	4
2	12V Power Adapter	1
3	BCL-ASK EXPT Manual	1

FUNCTIONAL BLOCK



ASK MODULATION / DEMODULATION

INTRODUCTION

“BCL-ASK” initiates the user to the various Carrier Modulation Techniques, normally adopted in practice

“BCL-ASK”: AMPLITUDE SHIFT KEYING MODULATOR / DEMODULATOR KIT.

FEATURES:

- 1) The trainer comes with an onboard data simulator, which generates the NRZ-L pattern depending on positions of 8-bit switch and the reference clock (256 KHz), which enables the trainers to work in a stand-alone mode.
- 2) Carrier modulation options are available on the kit.
 - a) Amplitude Shift Keying (ASK)
- 3) Onboard carrier generation circuit generates sine waves synchronized to the transmitted data. The three carrier sine waves generated onboard are of frequencies 1 MHz.
- 4) Interconnection facilities:
Sockets and patch chords are provided for on board connections.
- 5) Test Points:
All relevant test points are brought-out for observations. Observations are carried out on an oscilloscope.
- 6) Power Supply:
-12V, +12V, GND.

PRINCIPLES OF DIGITAL MODULATION AND DEMODULATION TECHNIQUES:

Digital Communication is the technique by which the whole information is represented in terms of binary digits, i.e., 'ones' and 'zeros'. These digits are represented by discrete voltage levels and the clock frequency of a digital communication scheme is generally low. For long distance transmission, the data is made to modulate a continuous wave (sine wave) carrier. These techniques are called Carrier Modulation Techniques. The various types of Carrier Modulation Techniques normally adopted in practice fall under three broad categories.

- a) Amplitude Shift Keying (ASK),
- b) Frequency Shift Keying (FSK),
- c) Phase Shift Keying (PSK).

a) AMPLITUDE SHIFT KEYING (ASK):

For all types of Carrier Modulation, the Carrier frequency should be at least 2 times base band of the modulating signal.

Assuming that the modulated carrier is a sine wave represented by the following equation.

$$C(t) = A(t) \cos wt$$

Where $C(t)$ = Carrier sine wave.
 $A(t)$ = Time varying amplitude
 wt = Time varying angle.

Now in amplitude shift keying, the carrier is being transmitted only when the modulating data is 'one' and when the data is 'zero' the carrier is rejected from transmission. Thus the resulting modulated output of this type of modulation can be represented as follows:

$$M(t) = r(t) * C(t)$$

Where $M(t)$ = Modulated Carrier
 $r(t)$ = Time varying modulating data which is either 'one' or 'zero'

Now $M(t) = \begin{matrix} A(t) \cos wt & \text{when modulating data is one.} \\ 0 & \text{when modulating data is zero.} \end{matrix}$

An Envelope detector is used to recover the data from the modulated carrier.

**EXPERIMENT
NO. 1**

EXPERIMENT NO 1

Amplitude Shift Keying Modulation and Demodulation

Aim:

To study the Amplitude Shift Keying Modulation and Demodulation.

Equipment:

ASK board, power supply, patch chord, Dual channel oscilloscope/DSO.

Theory:

Amplitude-shift keying (ASK) is a form of amplitude modulation that represents digital data as variations in the amplitude of a carrier wave. In an ASK system, the binary symbol 1 is represented by transmitting a fixed-amplitude carrier wave and fixed frequency for a bit duration of T seconds. If the signal value is 1 then the carrier signal will be transmitted; otherwise, a signal value of 0 will be transmitted.

Any digital modulation scheme uses a finite number of distinct signals to represent digital data. ASK uses a finite number of amplitudes, each assigned a unique pattern of binary digits. Usually, each amplitude encodes an equal number of bits. Each pattern of bits forms the symbol that is represented by the particular amplitude. The demodulator, which is designed specifically for the symbol-set used by the modulator, determines the amplitude of the received signal and maps it back to the symbol it represents, thus recovering the original data. Frequency and phase of the carrier are kept constant.

Like AM, ASK is also linear and sensitive to atmospheric noise, distortions, propagation conditions on different routes in PSTN, etc. Both ASK modulation and demodulation processes are relatively inexpensive. The ASK technique is also commonly used to transmit digital data over optical fiber. For LED transmitters, binary 1 is represented by a short pulse of light and binary 0 by the absence of light. Laser transmitters normally have a fixed "bias" current that causes the device to emit a low light level. This low level represents binary 0, while a higher-amplitude light wave represents binary 1.

The simplest and most common form of ASK operates as a switch, using the presence of a carrier wave to indicate a binary one and its absence to indicate a binary zero. This type of modulation is called on-off keying (OOK), and is used at radio frequencies to transmit Morse code (referred to as continuous wave operation)

More sophisticated encoding schemes have been developed which represent data in groups using additional amplitude levels. For instance, a four-level encoding scheme can represent two bits with each shift in amplitude; an eight-level scheme can represent three bits; and so on. These forms of amplitude-shift keying require a high signal-to-noise ratio for their recovery, as by their nature much of the signal is transmitted at reduced power.

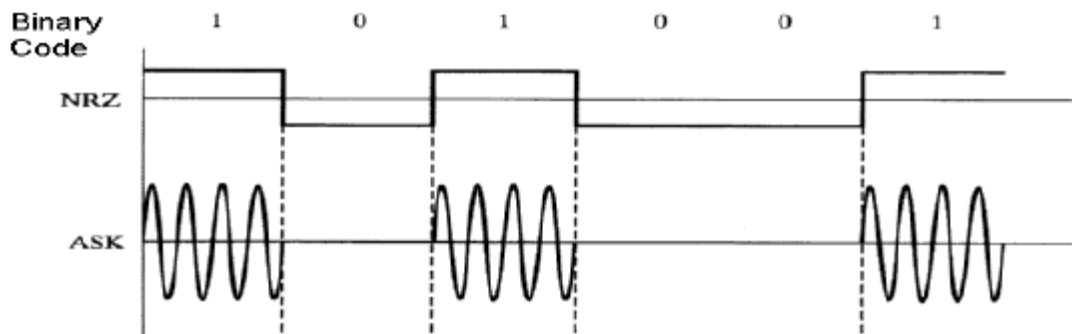


Fig 2. ASK waveforms

Procedure:

1. Connect power supply to kit with proper polarity and switch it ON.
2. Refer to KIT and carry out connections as mentioned below.
3. Observe carrier frequency at SINEWAVE post and note down the same.
4. Set data pattern as 10111001 using switch SW1 at DATA post.
5. Connect SINEWAVE to IN1 post.
6. Short IN2 post to ground using patch chord.
7. Connect DATA post to IN3 post.
8. Observe ASK modulated signal at OUT1 post. Compare it with DATA. It is similar to fig 2 shown above.
9. To demodulate this signal connect OUT1 post to IN4 post.
10. ASK demodulator uses envelope detector technique to demodulate the ASK signal.
11. Observe recovered data pattern at OUT2 post. Compare it with DATA.

Waveforms:

